

Mesh Mortal Kombat: Real-time voxelized soft-body destruction

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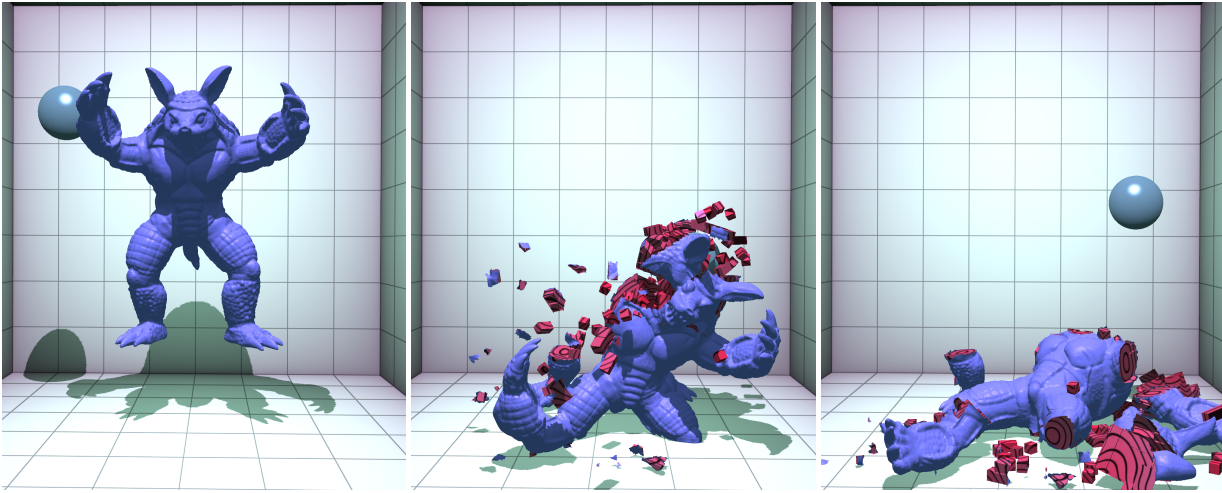


Figure 1: Sequence showing armadillo mesh being destroyed.

ABSTRACT

Mesh Mortal Kombat uses a new non-physical voxel constraint based on Gram-Schmidt orthonormalization within a position-based dynamics (PBD) framework. After the input mesh is voxelized, we create 8 PBD particles which are constrained to approximate each voxel. Breakable face-to-face constraints between voxels, which are also based on Gram-Schmidt, are created for pairs of neighboring voxels. The regular voxel grid allows efficient parallel processing in a compute shader using Gauss-Seidel iterations with only 1 voxel constraint partition and 3 face constraint partitions. This allows us to minimize the number of compute shader dispatches and memory barriers per frame. Destruction is achieved by disabling face-to-face voxel constraints when the strain limit is exceeded. Anisotropic materials can be created by setting different strain limits on x-, y-, and z- faces. This allows meshes to be peeled into sheets, shredded into strips, or fractured into voxels.

CCS CONCEPTS

• Computing methodologies → Real-time simulation.

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KEYWORDS

animation, simulation, physics, fracture

ACM Reference Format:

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1 INTRODUCTION

Mesh Mortal Kombat is a real time soft-body destruction demonstration based on position based dynamics [Müller et al. 2007] and some new constraints we developed based on voxels. Our demo is implemented using compute shaders, which are available on every modern graphics API, hardware vendor and OS - even in the web browser with WebGPU.

We begin by voxelizing the input mesh into a low resolution grid, up to $32 \times 32 \times 32$ in this demo. For each voxel, 8 PBD particles are created. These represent the mass, position and velocity of the voxel. The input mesh is embedded into the surface voxels which then act as an animation cage for the mesh.

2 VOXEL CONSTRAINTS

Our constraint is based on Gram-Schmidt orthonormalization and it keeps the voxels in a cubical configuration. Gram-Schmidt is a classical numerical technique for generating a set of orthonormal vectors from a set of input vectors. Bias in the Gram-Schmidt results

is a known issue [Müller et al. 2005], but our approach overcomes this limitation.

We store the particles in Morton code order which allows our voxels to be stateless - they occupy no additional memory beyond that required to store the PBD particles. The resulting constraint is similar to shape matching constraints [Müller et al. 2005, 2022] used in PBD, except we don't need to explicitly store the rest pose shape, or a matrix or quaternion representing the orientation of the shape.

A compelling benefit of the PBD framework is that rotational properties like angular momentum and moment of inertia emerge from the process of solving constraints. So we don't need to explicitly calculate or store these quantities.

3 FACE-TO-FACE CONSTRAINTS

We also use Gram-Schmidt to define voxel face constraints that glue adjoining voxels together. The emergent rotational properties automatically update when these constraints are broken. Each piece of the broken model will exhibit plausible rotational behavior.

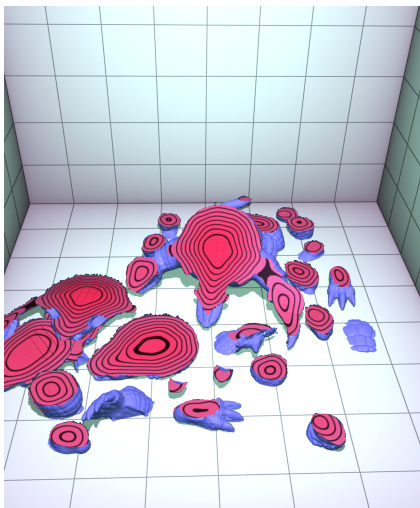


Figure 2: Planar anisotropy results in models that break into slabs.

Avoiding read-write conflicts is a critical concern in parallel programming. In this case these conflicts can arise when multiple constraints being solved in parallel try to update the same particle. This problem is avoided by grouping constraints into different partitions so that no particle is influenced by more than one constraint in the partition. Minimizing the number of these partitions reduces the number of compute shader dispatches and memory barrier calls and improves performance. Our voxel and face constraints are grouped into a total of 4 partitions.

The strain, or stretch, of the constraint determines when it should be broken. Defining the constraints on faces allows us to achieve direction-dependent or anisotropic destruction effects. So we can model materials that shatter into loose voxels, delaminate into layers (see Figure 2) or shred into strips (see Figure 3). We randomize our strain limits to give each destroyed model some natural variation.

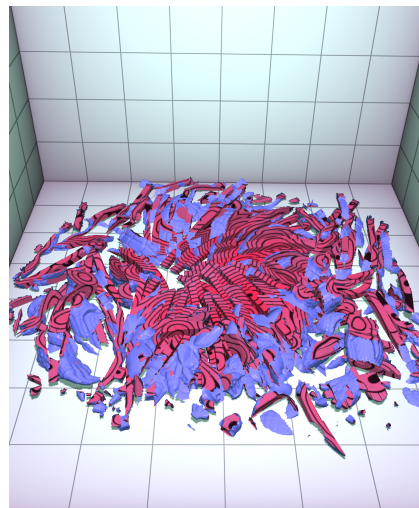


Figure 3: Linear anisotropy encourages models to fracture into strips

The 3d voxel grid also allows us to use classical morphological operators to compute a skeleton and distance transform, which can be used to set physical properties and generate internal texture details. This allows us to model additional inhomogeneous structures, by making the skeleton or skin surface stronger than nearby materials.

4 RENDERING

We use task and mesh shaders for efficient rendering of the deformed mesh, creating one surface meshlet and one interior meshlet per voxel. Inner voxel surfaces are culled in the task shader unless the corresponding face constraints between voxels are broken. So there is no rendering overhead for unbroken constraints.

In the demo, we show isocontours of the distance transform on the interior pieces, but the skeleton, or 3d texture coordinates could also be used to synthesize interior details, or a preauthored 3d texture could be used.

ACKNOWLEDGMENTS

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